

Business Recommendations

Purpose

This document translates the actual customer_churn.csv analysis into practical retention actions. Recommendations are based on observed segment profiles, churn rates, model metrics, and predictive features. Feature importance indicates predictive association and does not prove causation.

Dataset evidence

The dataset contains 500 customers and 53 observed churn cases (10.60%). There are no missing values and no duplicate rows. CustomerID was used only for traceability, never as a machine-learning feature.

Executive conclusion

Segment 2, Highest Churn Priority, needs the highest retention attention with 19.59% churn across 97 customers. Segment 4, Lowest Churn Stability, is the most stable at 2.25%. Segment 2 has the highest average TotalCharges (4490.11) and should be treated as a value-protection priority, while recognizing that TotalCharges is only a value proxy.

Recommended operating principle

Use the highest-churn segment for immediate retention experiments, use the stable segments as a control or loyalty population, and review model probabilities together with customer context before contacting customers.

Segment 0: Higher-Value Premium Base

Priority

HIGH - above overall churn rate

Observed profile

75 customers (15.00% of the dataset); average tenure 37.13; average MonthlyCharges 122.71; average TotalCharges 4264.80; senior citizen share 44.00%; observed churn 12.00%.

Observed categorical pattern

Contract distribution: {"Month-To-Month": 29.33, "One Year": 45.33, "Two Year": 25.33}.

Payment distribution: {"Electronic Check": 100.0}. Paperless billing distribution: {"Yes": 100.0}. These are descriptive patterns, not causal explanations.

Model evidence

Accuracy 1.000; precision 1.000; recall 1.000; F1 1.000; ROC-AUC 1.000. Top predictive features: ["Tenure", "TotalCharges", "MonthlyCharges", "Contract_Two Year", "SeniorCitizen"]. Low recall or zero positive predictions means the model should not be used as an automatic rejection or contact decision.

Recommended strategy

Use targeted outreach and test contract, billing, and engagement interventions. Compare outcomes against a holdout group before scaling the campaign.

Actions

1) Define a contact list using segment membership and model probability. 2) Test one intervention at a time. 3) Record offer, contact channel, response, and subsequent churn. 4) Review results by segment and by churn outcome.

Success measures

Primary: reduction in churn among contacted eligible customers. Secondary: save rate, offer acceptance, complaint rate, incremental revenue, and cost per retained customer. Always compare with a holdout group where permitted.

Segment 1: Established Customer Base 1

Priority

MEDIUM - monitor and nurture

Observed profile

86 customers (17.20% of the dataset); average tenure 36.71; average MonthlyCharges 112.20; average TotalCharges 4190.29; senior citizen share 50.00%; observed churn 4.65%.

Observed categorical pattern

Contract distribution: {"Month-To-Month": 26.74, "One Year": 44.19, "Two Year": 29.07}.

Payment distribution: {"Bank Transfer": 100.0}. Paperless billing distribution: {"Yes": 100.0}. These are descriptive patterns, not causal explanations.

Model evidence

Accuracy 0.944; precision 0.000; recall 0.000; F1 0.000; ROC-AUC 1.000. Top predictive features: ["Tenure", "TotalCharges", "MonthlyCharges", "SeniorCitizen", "Contract_One Year"]. Low recall or zero positive predictions means the model should not be used as an automatic rejection or contact decision.

Recommended strategy

Use personalized engagement, periodic satisfaction checks, and model-score monitoring. Escalate customers whose predicted risk increases rather than applying broad discounts.

Actions

1) Define a contact list using segment membership and model probability. 2) Test one intervention at a time. 3) Record offer, contact channel, response, and subsequent churn. 4) Review results by segment and by churn outcome.

Success measures

Primary: reduction in churn among contacted eligible customers. Secondary: save rate, offer acceptance, complaint rate, incremental revenue, and cost per retained customer. Always compare with a holdout group where permitted.

Segment 2: Highest Churn Priority

Priority

CRITICAL - first retention priority

Observed profile

97 customers (19.40% of the dataset); average tenure 37.40; average MonthlyCharges 112.58; average TotalCharges 4490.11; senior citizen share 56.70%; observed churn 19.59%.

Observed categorical pattern

Contract distribution: {"Month-To-Month": 100.0}. Payment distribution: {"Bank Transfer": 28.87, "Credit Card": 38.14, "Electronic Check": 32.99}. Paperless billing distribution: {"No": 100.0}. These are descriptive patterns, not causal explanations.

Model evidence

Accuracy 1.000; precision 1.000; recall 1.000; F1 1.000; ROC-AUC 1.000. Top predictive features: ["Tenure", "MonthlyCharges", "TotalCharges", "SeniorCitizen", "PaymentMethod_Electronic Check"]. Low recall or zero positive predictions means the model should not be used as an automatic rejection or contact decision.

Recommended strategy

Launch a focused save campaign. Review contract renewal options, payment experience, billing clarity, and service friction. Offer a measured contract incentive only after identifying the relevant friction; do not assume price is the cause.

Actions

1) Define a contact list using segment membership and model probability. 2) Test one intervention at a time. 3) Record offer, contact channel, response, and subsequent churn. 4) Review results by segment and by churn outcome.

Success measures

Primary: reduction in churn among contacted eligible customers. Secondary: save rate, offer acceptance, complaint rate, incremental revenue, and cost per retained customer. Always compare with a holdout group where permitted.

Segment 3: Lower-Charge Opportunity

Priority

HIGH - above overall churn rate

Observed profile

82 customers (16.40% of the dataset); average tenure 37.29; average MonthlyCharges 108.83; average TotalCharges 4357.30; senior citizen share 50.00%; observed churn 17.07%.

Observed categorical pattern

Contract distribution: {"Month-To-Month": 34.15, "One Year": 30.49, "Two Year": 35.37}.

Payment distribution: {"Credit Card": 100.0}. Paperless billing distribution: {"Yes": 100.0}.

These are descriptive patterns, not causal explanations.

Model evidence

Accuracy 0.941; precision 1.000; recall 0.667; F1 0.800; ROC-AUC 1.000. Top predictive features: ["Tenure", "MonthlyCharges", "TotalCharges", "Contract_Month-To-Month", "Contract_One Year"]. Low recall or zero positive predictions means the model should not be used as an automatic rejection or contact decision.

Recommended strategy

Use targeted outreach and test contract, billing, and engagement interventions. Compare outcomes against a holdout group before scaling the campaign.

Actions

1) Define a contact list using segment membership and model probability. 2) Test one intervention at a time. 3) Record offer, contact channel, response, and subsequent churn. 4) Review results by segment and by churn outcome.

Success measures

Primary: reduction in churn among contacted eligible customers. Secondary: save rate, offer acceptance, complaint rate, incremental revenue, and cost per retained customer. Always compare with a holdout group where permitted.

Segment 4: Lowest Churn Stability

Priority

LOW - protect and learn

Observed profile

89 customers (17.80% of the dataset); average tenure 37.47; average MonthlyCharges 110.46; average TotalCharges 3761.88; senior citizen share 47.19%; observed churn 2.25%.

Observed categorical pattern

Contract distribution: {"One Year": 100.0}. Payment distribution: {"Bank Transfer": 30.34, "Credit Card": 38.2, "Electronic Check": 31.46}. Paperless billing distribution: {"No": 100.0}. These are descriptive patterns, not causal explanations.

Model evidence

Accuracy 0.944; precision 0.000; recall 0.000; F1 0.000; ROC-AUC not valid. Top predictive features: ["Tenure", "TotalCharges", "MonthlyCharges", "SeniorCitizen", "PaymentMethod_Credit Card"]. Low recall or zero positive predictions means the model should not be used as an automatic rejection or contact decision.

Recommended strategy

Maintain service quality, use lightweight loyalty recognition, and avoid unnecessary discounting. This segment can provide a useful comparison group when measuring retention experiments.

Actions

1) Define a contact list using segment membership and model probability. 2) Test one intervention at a time. 3) Record offer, contact channel, response, and subsequent churn. 4) Review results by segment and by churn outcome.

Success measures

Primary: reduction in churn among contacted eligible customers. Secondary: save rate, offer acceptance, complaint rate, incremental revenue, and cost per retained customer. Always compare with a holdout group where permitted.

Segment 5: Established Customer Base 2

Priority

MEDIUM - monitor and nurture

Observed profile

71 customers (14.20% of the dataset); average tenure 32.44; average MonthlyCharges 116.77; average TotalCharges 4381.25; senior citizen share 49.30%; observed churn 7.04%.

Observed categorical pattern

Contract distribution: {"Two Year": 100.0}. Payment distribution: {"Bank Transfer": 25.35, "Credit Card": 35.21, "Electronic Check": 39.44}. Paperless billing distribution: {"No": 100.0}. These are descriptive patterns, not causal explanations.

Model evidence

Accuracy 0.933; precision 0.000; recall 0.000; F1 0.000; ROC-AUC 0.786. Top predictive features: ["Tenure", "MonthlyCharges", "TotalCharges", "PaymentMethod_Bank Transfer", "SeniorCitizen"]. Low recall or zero positive predictions means the model should not be used as an automatic rejection or contact decision.

Recommended strategy

Use personalized engagement, periodic satisfaction checks, and model-score monitoring. Escalate customers whose predicted risk increases rather than applying broad discounts.

Actions

1) Define a contact list using segment membership and model probability. 2) Test one intervention at a time. 3) Record offer, contact channel, response, and subsequent churn. 4) Review results by segment and by churn outcome.

Success measures

Primary: reduction in churn among contacted eligible customers. Secondary: save rate, offer acceptance, complaint rate, incremental revenue, and cost per retained customer. Always compare with a holdout group where permitted.

Retention Program Design

Priority order

1. Highest Churn Priority at 19.59% churn. 2. Segments at or above the overall churn rate of 10.60%. 3. Highest Churn Priority for value protection. 4. Stable segments for loyalty and control-group learning.

Contact policy

Use a human-reviewed threshold for outreach. Do not treat a predicted probability as a guaranteed outcome. Suppress customers who recently received the same offer, have unresolved complaints, or are legally restricted from the campaign.

Experiment policy

For each segment, define a treatment group and a comparable holdout group. Predefine the success metric and evaluation window. Avoid changing multiple offers simultaneously because that makes the result difficult to interpret.

Data feedback loop

Store campaign exposure, offer type, channel, response, renewal, and churn outcome. Recalculate segment profiles periodically because customer composition and contract mix can change.

Governance

Review fairness and unintended impact, especially where SeniorCitizen appears among important predictive features. Use the model to support customer service decisions, not to deny service or impose punitive pricing.

Implementation Roadmap and Limitations

First 30 days

Validate the highest-churn segment with customer-service and billing teams. Create a small controlled retention campaign. Capture treatment, response, cost, and churn outcomes.

Days 31-60

Compare campaign and holdout results. Refine the contact threshold and messaging. Review false negatives and false positives, not accuracy alone.

Days 61-90

Scale only interventions with measurable incremental retention. Retrain the segment and prediction workflow when new labeled outcomes are available.

Model limitations

The dataset has 500 rows and only 53 churn cases. Segment-level test metrics may be unstable, and several models predict no positive class in their small test sets. ROC-AUC is omitted where mathematically invalid. Clustering depends on feature scaling and algorithm choices.

Final caution

The recommendations describe sensible actions suggested by observed data. They should be validated through controlled business experiments and customer feedback before becoming permanent policy.